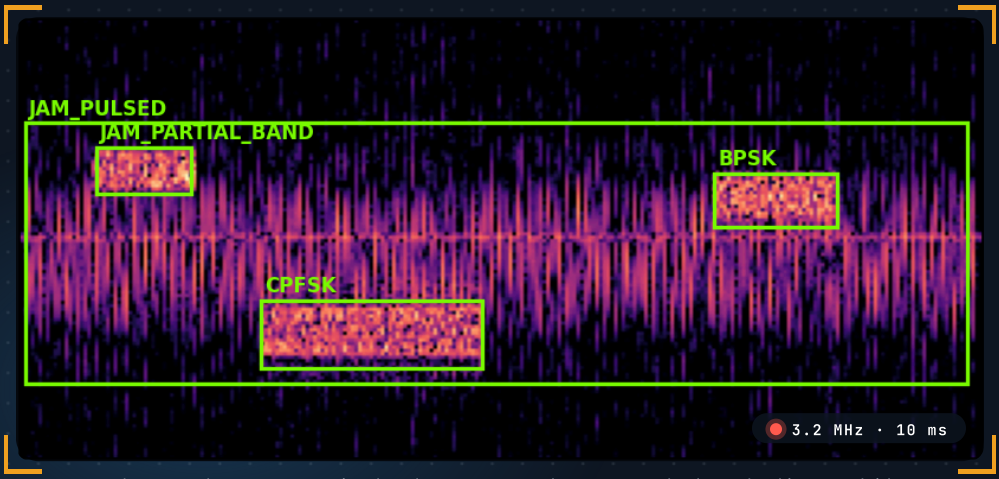
NEXA

SEDIC@FOI2026



NEXA's own boxes and answers on a simulated capture at 20 dB, not retouched: a pulsed jammer laid over three other signals, and every one still found and named.

TRACK I · RF SIGNAL INTELLIGENCE

Every emitter on the band, named.

NEXA watches a whole band of radio spectrum, finds every transmission in it and tells the operator what each one is: jammer, radar, voice or data. It runs offline on one workstation, and people stay in control.

HOW IT WORKS

1 Capture	2 Detect	3 Classify	4 Track	5 Resolve	6 Listen
Receiver, IQ file or simulator	A box on every emitter	60 classes, per window	Votes over time	Settles look-alikes	Voice and Morse to text

60	90%	79.4%	0
signal classes in 12 families	right signal family, 15 dB, 8-window vote	exact class out of 60, SNR ≥ 10 dB, 16-window vote	data leaves the machine

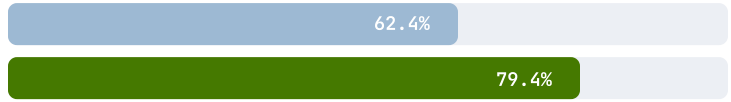
NEXA

SEDIC 26 Strategic Electronic Defence Innovation Challenge 2026

Named exactly, out of 60.

79.4% **exact class, per emitter**

All 60 classes · SNR ≥ 10 dB · vote over 16 windows (2.6 ms of signal) · fixed test set of 120,000 windows the model never trained on. Chance is 1.7%. Right family: 90%.



ASK	100%	FSK	100%	Radar	95%	QAM	94%
Jammers	93%	OFDM / LTE / NR	93%	PSK	91%	Analog voice	86%
Spread spectrum	82%	Pulse comms	70%	APSK	69%	Noise floor	66%

Right family · 15 dB · vote over 8 windows · 32 fresh tracks per class from our own generators, one emitter at a time.

Tri-Branch Deep Fusion Network for Multi-Class RF Signal Classification

Each 160 µs window is seen three ways by small neural networks, then fused with measured physics of the signal. Look-alike sizes (QAM, APSK, high-order PSK) are named as a group, and every answer carries a calibrated band: `HIGH` `MEDIUM` `UNCERTAIN`

raw IQ	*CNN*
spectrogram	*CNN*
spectrum	*CNN*
physics + expert	*features*

Built for contested spectrum

Sees the attack
Ten jammer types and eight radar waveforms, named next to the signal they sit on.

Says how sure it is
Uncertain answers go to a person, not into the log as fact.

Stays in the room
Detection, speech recognition and AI reading run on one offline machine.

It learns, but only with approval

1. **Operator corrects** — One click.
2. **Analyst approves** — A second person.
3. **Retrain** — Started by a person.
4. **Gate** — Goes live only if better.

NEXA

Group NEXA · Eavan Tan (leader), Eileen Yeoh Jing Han, Jessy Pang Xin Yuan, Chua Xin Ying · Advisor Dr Noureen Talpur
Universiti Teknologi PETRONAS · SEDIC@FOI2026

`sedic-ai-nexa.vercel.app`

Try it | Watch | Brochure